
Personal Tutor Allocation Tool

Staff User Guide - Version 1.0

Repository: https://github.com/KTAswathi/EDIA_PT_Allocation_Tool

Handover release: v1.0.0

Purpose

This guide is intended to be self-contained: a new staff user should be able to prepare the annual inputs, run the tool, interpret the output and complete the required checks without needing to understand or edit the Python code.

Annual data → Validate → Allocate → Improve EDIA grouping → Excel report → Staff review

Core principle

The tool produces a **proposed allocation**. It supports staff decision-making but does not replace staff review or professional judgement. Safeguarding always takes priority over EDIA grouping preferences.

Quick start

For an annual run:

1. Download the GitHub repository using **Code** → **Download ZIP** and unzip it.
2. Inside the unzipped folder, create **annual_inputs/<academic-year>/.**
3. Put the Accepted Applicants workbook, annual Tutor List and populated configuration CSVs there.
4. Open Terminal or PowerShell.
5. Type `cd` including the space, then drag the unzipped **EDIA_PT_Allocation_Tool** folder into the Terminal window and press Enter.
6. Run the command given in Section 5.
7. Open **annual_outputs/<academic-year>/PT_Allocation_Output.xlsx**.
8. Review all unallocated, manual-review and diagnostic cases before approving the allocation.

1. What the tool does

The tool combines annual student information, annual tutor availability and a small amount of staff-maintained configuration.

Automatically, it:

- validates required columns and values;
- checks that student numbers are unique;
- maps programme codes to staff-approved allocation streams;
- determines tutor eligibility from the annual Tutor List;
- ensures under-18 students are allocated only to DBS-checked tutors;
- creates capacity-aware baseline groups;
- improves EDIA-related grouping where possible without breaking mandatory rules;
- reports students who cannot be safely allocated;
- identifies warnings and manual-review cases; and
- produces a staff-facing Excel workbook.

It deliberately does not:

- infer support severity from disability information;
- decide whether a tutor is experienced or trusted;
- guess programme-to-allocation-stream mappings;
- automatically resolve returning/repeating student circumstances;
- optimise ethnicity or maintain an automated list of countries in conflict;
- publish allocations; or
- replace professional judgement.

2. Annual inputs

There are two main School/University files and four small configuration files.

Input	Required?	Purpose
Accepted Applicants workbook	Yes	Student cohort and demographic/entry information.
Annual Tutor List workbook	Yes	Tutor availability for each allocation stream.
programme_stream_mapping.csv	Yes	Maps each programme code to an allocation stream.
manual_tutors.csv	Yes	Annual tutor ID, DBS, experience and preferred capacity.
manual_students.csv	Optional	Approved support, entry-route and returning/repeating markers.
qualification_route_mapping.csv	Optional	Approved qualification-to-entry-route mappings.

2.1 Accepted Applicants workbook

The workbook must contain a sheet named **Report** and these headers:

Programme code	Programme	Student number	Sex
Student status	Country of domicile	Home / overseas	Age at entry
IFP Progressor?	Qualification name	Qualification level	

Key rules:

- **Student number** is the authoritative student ID and must be unique.
- **Age at entry** must be a whole number.
- Age is also used to derive mature status: age 21 or above is mature.
- A positive **IFP Progressor?** value sets entry route to IFP.
- **Home / overseas**, when populated, must be **Home** or **Overseas**.
- Disability information may be present but is **not** used to infer support severity.
- The current source does not reliably provide year of study; this remains blank and is reported as a warning.

2.2 Annual Tutor List

The workbook must contain a sheet named **Tutor List**, a **Tutor name** column and seven eligibility columns:

- CS year 1
- Maths and CS year 1
- CS with AI year 1
- CS with Innov year 1
- Computer Science MSc
- Cyber MSc
- Immersive Technologies MSc

Each eligibility value must be exactly:

1 = eligible 0 = not eligible

A tutor with all zeroes is ignored. Every tutor with at least one 1 must also appear in **manual_tutors.csv**.

Data protection

Real student data, populated annual configuration and allocation outputs must remain in an approved secure location. Do **not** commit live student data or annual operational files to GitHub.

3. Files staff maintain

3.1 Programme-to-stream mapping

File:

```
programme_stream_mapping.csv
```

Columns:

```
programme_code,allocation_stream
```

Example:

```
programme_code,allocation_stream
4COSC006UU,cs_year_1
4COSC019UU,maths_and_cs_year_1
4COSC018UU,maths_and_cs_year_1
```

```
4COMS002UU,cs_with_ai_year_1
4COMS001UU,cs_with_ai_year_1
```

Different programme codes, for example BSc and MEng variants, may map to the same allocation stream.

Current valid streams are:

- cs_year_1
- maths_and_cs_year_1
- cs_with_ai_year_1
- cs_with_innov_year_1
- computer_science_msc
- cyber_msc
- immersive_technologies_msc

If a new programme code belongs to one of these existing streams, simply add a row to the CSV. No Python change is required.

A genuinely new allocation category requires a maintained software change.

Programme mappings are explicit

Programme names are never used to guess a stream. If Accepted Applicants contains an unmapped programme code, the tool stops and asks staff to resolve it rather than silently guessing.

3.2 Annual tutor configuration

File:

```
manual_tutors.csv
```

Columns:

```
tutor_name,tutor_id,dbs_checked,experience_level,
preferred_capacity,admin_role
```

Column	Meaning
tutor_name	Must correspond to the Tutor List name.
tutor_id	Unique staff-maintained ID used in outputs.
dbs_checked	Must explicitly be True or False .
experience_level	standard or experienced ; manually maintained.
preferred_capacity	Positive whole number. Blank defaults to 8.
admin_role	Optional reference information; does not automatically alter capacity.

Preferred capacity is a soft target, not a hard maximum. For example, a tutor with capacity 4 should receive proportionally fewer students than a tutor with capacity 8, but groups may exceed the stated value where the cohort requires it.

DBS status must never be assumed.

The Tutor List currently has no stable staff ID used by the tool, so normalised tutor-name matching is the fallback. Names must therefore remain unique and consistent between the Tutor List and `manual_tutors.csv`.

3.3 Optional manual student information

File:

```
manual_students.csv
```

Columns:

```
student_number,support_need,entry_route,returning_student
```

This does **not** mean “manually allocated students”. It is simply optional extra information not reliably available in the main student report.

Support need:

Value	Meaning
0	none
1	standard
2	complex

Only explicitly approved support information should be entered. A disability code or description must not be converted automatically into a support level.

Allowed manual entry routes are:

A-level, BTEC, Access, Foundation, IFP, Postgraduate, Other

`returning_student` may be `True`, `False`, or blank. Returning/repeating students are displayed for manual review and do not automatically trigger reallocation.

3.4 Optional qualification mapping

File:

```
qualification_route_mapping.csv
```

Columns:

```
qualification_name,qualification_level,entry_route
```

Example:

```
BTEC National Extended Diploma,Level 3,BTEC
Access to HE Diploma,Level 3,Access
```

Unknown qualifications are not guessed. They remain blank and create a warning.

4. How the allocation works

The algorithm first creates an allocation satisfying all mandatory rules and balancing tutor loads according to preferred capacity. It then improves EDIA-related characteristics by moving/swapping students where this improves the diagnostic objective without breaking mandatory rules.

4.1 Hard constraints

These may never be deliberately violated:

1. **One allocation:** every hard-feasible student is allocated exactly once.
2. **Tutor eligibility:** students may only go to tutors eligible for their allocation stream.
3. **Under-18 safeguarding:** every student under 18 must have a DBS-checked tutor.
4. **Impossible cases:** if no valid assignment exists, the student remains unallocated and the reason is reported.

4.2 Soft EDIA considerations

Where possible, the tool improves:

- gender isolation;
- Home/Overseas isolation;
- concentration of Overseas students;
- repeated concentration of students from the same Overseas country;
- mature-student isolation;
- pathway/entry-route isolation;
- distribution of approved support markers;
- matching complex-support students to experienced tutors where the relevant approved information exists; and
- capacity balance.

When trade-offs are unavoidable, the principle is:

Safeguarding → reduce isolation → support transition to HE → reasonable overall balance

Reasonable

The optimiser does not claim that a mathematically perfect grouping always exists. Warnings may remain because of cohort composition, tutor availability or mandatory safeguarding constraints.

5. Running the tool

5.1 Recommended annual folders

After downloading and unzipping the repository, create:

```
EDIA_PT_Allocation_Tool/
|
```

```

|-- src/                <- do not edit
|-- config/             <- blank templates
|
|-- annual_inputs/
|   '-- 2026-27/
|       |-- AcceptedApplicants.xlsx
|       |-- Tutor List 26-27.xlsx
|       |-- programme_stream_mapping.csv
|       |-- manual_tutors.csv
|       |-- manual_students.csv
|       '-- qualification_route_mapping.csv
|
|-- annual_outputs/

```

The CSV templates in `config/` can be copied into the relevant annual input folder and populated.

5.2 Opening Terminal in the correct folder

For a user unfamiliar with Terminal:

1. Open Terminal on Mac, or PowerShell/Command Prompt on Windows.
2. Type:

```
cd
```

including the space after `cd`.

3. Drag the unzipped `EDIA_PT_Allocation_Tool` folder into the Terminal window.
4. Press Enter.

Terminal is now working inside the tool's main folder.

5.3 Run command

For academic year 2026-27:

```

python src/run_allocation.py \
"annual_inputs/2026-27/AcceptedApplicants.xlsx" \
"annual_inputs/2026-27/Tutor List 26-27.xlsx" \
--programme-map \
"annual_inputs/2026-27/programme_stream_mapping.csv" \
--tutor-config \
"annual_inputs/2026-27/manual_tutors.csv" \
--student-config \
"annual_inputs/2026-27/manual_students.csv" \
--qualification-map \
"annual_inputs/2026-27/qualification_route_mapping.csv" \
--output-dir "annual_outputs/2026-27"

```

If `manual_students.csv` or `qualification_route_mapping.csv` are not being used, their corresponding options may be omitted.

The program automatically creates the requested output folder if it does not already exist.

5.4 Exit codes

Code	Meaning	Action
0	Successful run	Open the workbook and complete staff review.
1	Run completed but one or more students could not be safely allocated	Review Unallocated ; staff intervention is required.
2	Invalid or incomplete input/configuration	Correct the indicated problem and rerun.

Exit code 1 is not a software failure

For example, an under-18 student with no eligible DBS-checked tutor should remain unallocated. The program must not violate safeguarding merely to produce a complete spreadsheet.

6. Understanding the output

The main staff-facing result is:

PT_Allocation_Output.xlsx

It is automatically saved inside the chosen annual output folder.

Worksheet	Purpose
Run Summary	Students supplied, allocated/unallocated, group counts and warning totals.
Proposed Allocations	One row per allocated student, showing programme, stream, tutor and group.
Group Summary	One row per tutor group with size, preferred capacity and composition.
Manual Review	Returning/repeating students and relevant adapter warnings.
EDIA Diagnostics	Complete group-level diagnostic information.
Unallocated	Students for whom no valid hard-constraint allocation exists, with the reason.

Recommended review order

Run Summary → Unallocated → Manual Review → Group Summary → EDIA Diagnostics → Proposed Allocations

Before approving the allocation, check:

- every expected student appears;
- all unallocated cases are understood and resolved;
- returning/repeating cases have been reviewed;
- remaining EDIA warnings are reasonable for the cohort;
- group sizes are operationally reasonable;
- relevant workload or pastoral circumstances are represented; and
- any manual amendments are appropriate.

Manual edits are not automatically revalidated

If staff edit the final workbook manually, the changes do not pass back through the algorithm. Any manual reassignment must therefore be checked again for tutor-stream eligibility and, for under-18 students, DBS safeguarding.

The output directory also contains:

```
students.csv, tutors.csv, baseline_allocations.csv, final_allocations.csv, diagnostics.csv,
warnings.txt, unallocated.csv, run_summary.txt
```

These are retained mainly for transparency, checking and troubleshooting.

7. Troubleshooting

Problem	Meaning	Action
Unknown programme code	Programme is not mapped.	Add its exact code to <code>programme_stream_mapping.csv</code> .
Tutor missing configuration	Eligible Tutor List tutor has no configuration row.	Add the tutor to <code>manual_tutors.csv</code> .
Missing DBS value	Safeguarding status is unknown.	Confirm it and enter exactly True or False .
Duplicate student number	Student IDs are not unique.	Correct the applicant input.
Empty applicant report	No applicant records were found.	Obtain/download the populated Accepted Applicants report.
Unknown qualification	Entry route cannot safely be inferred.	Add an approved mapping or leave blank for review.
No eligible DBS tutor	An under-18 student has no safe eligible assignment.	Leave unallocated and resolve manually.

8. Known limitations and first live use

Version 1 has the following known limitations:

- the EDIA optimiser improves the allocation but does not guarantee a mathematical global optimum;
- Overseas fee status is an acknowledged imperfect proxy for international-student status;
- tutor experience is manually maintained;
- support severity must be supplied explicitly and is never inferred from disability information;
- tutor matching currently falls back to normalised names because the Tutor List does not contain a stable staff identifier used by the tool;
- returning/repeating students remain manual-review cases;

- a genuinely new allocation stream requires a maintained software change; and
- the first populated live Accepted Applicants export should be checked before the tool is treated as fully operational.

Testing completed before handover

Validation evidence

The full development repository passed **70 tests with 0 failures**.

The clean handover repository passed **56 production-relevant tests with 0 failures**.

A final end-to-end smoke test used:

- 601 fictional students;
- 21 fictional tutors;
- all seven allocation streams;
- 600 feasible allocations; and
- one deliberately impossible under-18 case.

The impossible case was correctly left unallocated rather than violating the DBS requirement.

First populated live run

Treat the first run with a real populated Accepted Applicants export as a validation run.

Before relying on the result, compare a sample of source rows against the cleaned/output data and verify:

- student numbers;
- programme codes and allocation streams;
- age;
- Home/Overseas status;
- tutor identities and eligibility;
- DBS status;
- total students supplied/allocated/unallocated; and
- final group counts.

9. Annual checklist

Download current Accepted Applicants report.
Update programme mappings.
Update tutor experience/capacity.
Review qualification mappings.
Check **Unallocated**.
Review group sizes and EDIA diagnostics.
Apply professional judgement and approve.

Obtain current Tutor List.
Confirm DBS for every active tutor.
Add approved optional student information.
Run the allocation tool.
Check **Manual Review**.
Review proposed allocations.
Keep live inputs/outputs out of GitHub.

Where to find more detail

This Staff User Guide is intended to contain everything needed for normal annual use.

The repository's **README.md** provides the fuller technical and operational reference, while **RULE_SPEC.md** records the detailed allocation rules implemented by the software.

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End of Staff User Guide